

3.4. SELECTION BIAS

Sample selection bias results from the tendency of returns only to be observed when underlying asset values are high.

Buildings tend to be sold when their values are high—otherwise, many sellers postpone sales until property values recover. This causes more transactions to be observed when the underlying real estate values are high.

In private equity, selection bias is acute. In buyout funds, companies are taken public only when stock values are high. Many venture capital investments are structured over multiple rounds. Better-performing companies tend to raise more money in more rounds. The venture capitalist tends to sell a small company, and the transaction is recorded, when the company's value is high. Distressed companies are usually not formally liquidated, and these “zombie” companies are often left as shell companies. When observing old companies without recent transactions, it is not clear whether these companies are alive and well or whether they are zombies.

To illustrate the selection bias problem, consider Figure 13.4, which is adapted from Korteweg and Sorensen (2010). Panel A shows the full universe of returns of an illiquid asset marked by dots. These returns (on the y-axis) are plotted contemporaneous with market returns (on the x-axis). In the full universe, there is no alpha, and the intercept of the line summarizing the relationship between the illiquid asset and the market goes through the origin (this line is called the security market line (SML; see chapter 6). The slope of the SML is the beta of the illiquid asset and is a measure of risk.

Panel B illustrates the sample selection problem. Bad returns, which are shaded gray, are not observed in the databases—we record transactions only when prices are high. Now only the black dots are reported. An estimated SML fitted to these observed returns yields a positive alpha when the true alpha is zero. The slope of the fitted SML is flatter than the slope of the true SML in panel A, and hence we underestimate beta. When we compute the volatility of the observed returns, we only count those returns that are high, and so the volatility estimate is biased downward. Thus we overestimate expected return, and we underestimate risk as measured by beta and volatility.

The statistical methodology for addressing selection bias was developed by James Heckman (1979), who won the Nobel Prize in 2000 for inventing these and other econometric techniques. Studies that use models to correct for these biases do not take such an extreme view as Figure 13.4 they allow the threshold above which returns are observed to vary over time and depend on company or property-level characteristics.¹⁹ The model of risk is sometimes extended to multifactor

¹⁹ See also Cochrane (2005) for selection bias models applied to venture capital and Fisher et al. (2003) for real estate. Korteweg, Kräussl, and Verwijmeren (2012) find that correcting for selection bias decreases the Sharpe ratio of art from 0.4 to 0.1.